



How to Configure DNS over HTTPS (DoH) Using DNSCrypt-Proxy in OPNsense

👤 [Dustin Casto](#) 📅 December 24, 2019 📖 [Routers](#) 💎 [DNS](#) , [OPNsense](#) , [Cloudflare](#)



Photo by [Simon](#) from [Pixabay](#)

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Historically, DNS is a service that was designed to be unencrypted. Whenever a device from your network is trying to go to a web address, it needs to determine the IP address of the website in order to access it. With the increasing levels of tracking and data sharing/selling, a growing awareness that having DNS traffic unencrypted is not a good idea from a privacy and security standpoint. ISPs and other entities are able to know which sites you visit even if all of your web traffic is encrypted.

Lately there has been a growing desire to encrypt DNS traffic as much as possible – an attempt to escape from some of the madness of increased data collection and tracking. There are a number of ways DNS can be encrypted and there are various pros/cons to each. DNS over HTTPS (DoH) is quickly becoming a popular way to encrypt DNS traffic. Instead of sending

DNS traffic on UDP port 53, it is sent over TCP port 443 just like all other encrypted web traffic. The DNS server has to support DoH in order for the DNS lookup to success.

Install the DNSCrypt-Proxy Plugin in OPNsense

When I first looked into configuring DoH in OPNsense, I saw some forum posts about how to do this by manually installing packages via SSH command line and tweaking the configuration. Since then, there is now a DNSCrypt-proxy plugin that can be installed which simplifies the process by exposing the configuration in the web GUI of OPNsense.

To install DNSCrypt-proxy in OPNsense, go to System > Firmware > Plugins. Click the “+” icon beside the os-dnscrypt-proxy plugin to begin the installation. There should be a new menu option under “Services” for “DNSCrypt-Proxy”.

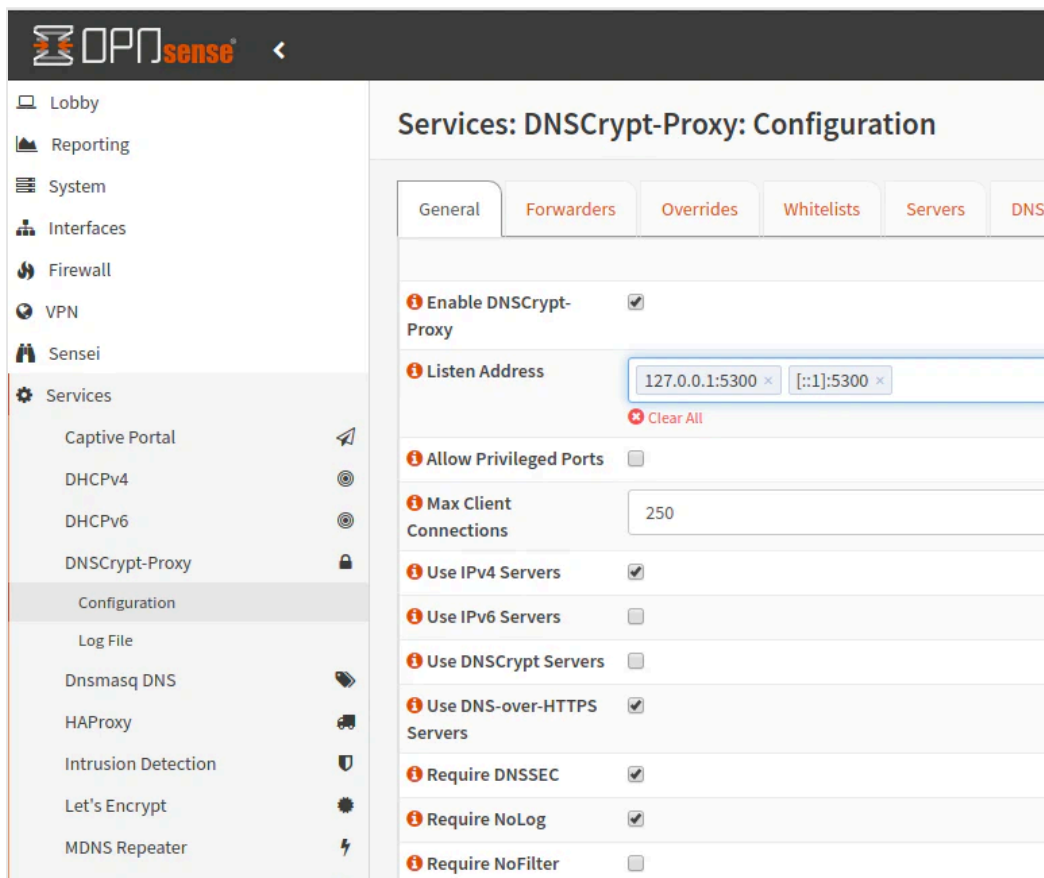
Configuring DNSCrypt-Proxy

Go to the Services > DNSCrypt-Proxy > Configuration page to begin configuring DNSCrypt-proxy. You will need to check “Enable DNSCrypt-Proxy” to enable the service. For the “Listen Address”, it will be defaulted to 127.0.0.1:5353 for IPv4 DNS queries and [::1]:5353 for IPv6 queries (remember that [::1] in IPv6 is equivalent to localhost 127.0.0.1 in IPv4). If you wish to use the default Unbound DNS service in OPNsense, leaving these values at the default is ok.

Warning: If you are also using the multicast DNS (MDNS-Repeater) plugin in OPNsense, you will need to change the default DNSCrypt-proxy port of 5353 to something else. I am using 5300 to keep it similar to the unencrypted port 53 of DNS. It is likely they chose 5353 as the default since it is similar to 53, but they likely did not think about the conflicts when enabling the multicast DNS plugin (multicast DNS is designed to use port 5353).

If you are planning to exclusively use DNSCrypt-proxy, you could check the “Allow Privileged Ports” box and then change the “Listen Address” to use

port 53. However, in my example I am using Unbound DNS so using DNSCrypt-proxy on port 53 is beyond the scope of this how-to. If you are not using IPv6, you could uncheck “Use IPv6 Servers” box. The next 5 checkboxes (“Use DNSCrypt Servers”, “Use DNS-over-HTTPS”, “Require DNSSEC”, “Require NoLog”, and “Require NoFilter”) I have checked except for “Use DNSCrypt Servers”. Since I am going to later choose specific DNS servers, none of these settings matter as stated in the [DNSCrypt-proxy documentation](#). Those options are the filters used when you allow DNSCrypt-proxy to randomly choose the fastest DNS servers to allow you have control over the types of services you wish to use.



The “Fallback Resolver” value defaults to 9.9.9.9:53 which is for the [Quad9](#) DNS service. You may change it to 1.1.1.1 if you prefer to use Cloudflare. This setting is used to obtain the IP addresses of upstream DNS resolvers in case the normal system DNS is not functioning properly. It is not used to resolve any other domain names that you wish to have encrypted. You may check the “Block IPv6” box if you are not using IPv6. The help information says that there could be a slight performance gain by disabling this option. I suppose that is due to the fact that it does not have to query any IPv6 servers when it is not necessary. It immediately returns a blank response. You may tweak any of the cache settings, but I left mine at the default settings. Finally, there is the “Server List” which allows you to manually

specify servers rather than allow a random DNS server selection. I personally feel more comfortable knowing which servers my DNS queries are being performed. I picked Cloudflare and Quad9 since they seem to be trusted, well-known DNS providers. Note that you have to use the name of the server and not the IP addresses. The list of servers can be found on the [DNSCrypt page](#).

The screenshot displays the DNSCrypt-proxy configuration web interface. On the left, a sidebar lists various services, with 'DNSCrypt-Proxy' highlighted. The main configuration area on the right includes several settings: 'Ephemeral Keys' (unchecked), 'TLS Disable Session Tickets' (unchecked), 'Fallback Resolver' (1.1.1.1:53), 'Block IPv6' (checked), 'Cache' (checked), 'Cache Size' (512), 'Cache Min TTL' (600), 'Cache Max TTL' (86400), 'Cache Negative Min TTL' (60), 'Cache Negative Max TTL' (600), and 'Server List' (cloudflare, quad9-doh-ip4-filter-pri). A 'Save' button is located at the bottom of the configuration panel.

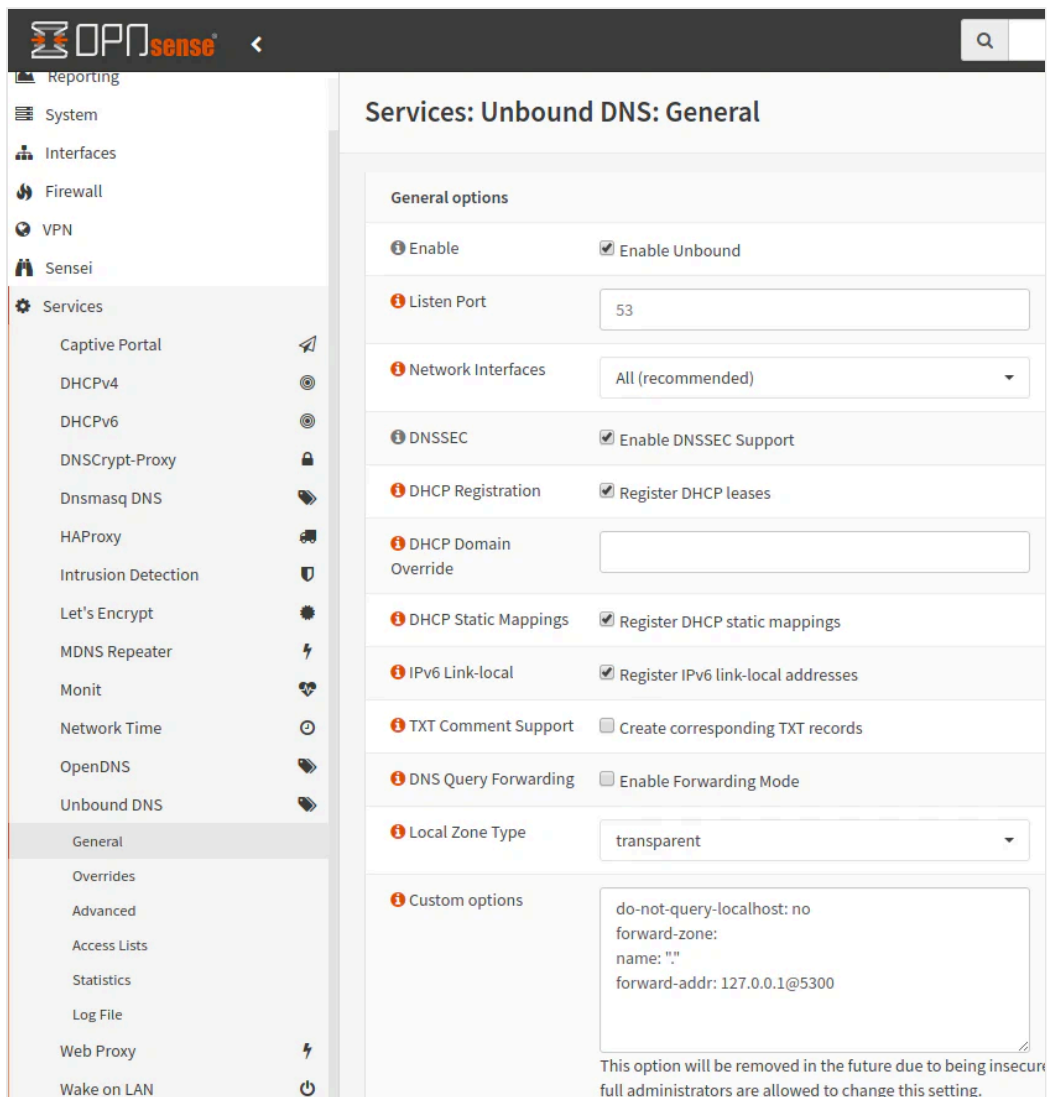
Configuring Unbound DNS (Optional)

If you wish to use Unbound DNS along with DNSCrypt-proxy, this is entirely possible. The main reason I chose to keep using Unbound DNS is that I have it set up to resolve hostnames on my local network. Unbound DNS also automatically advertises itself as the DNS server via DHCP for any LAN/VLAN networks you have set up to use DHCP. I do not know if DNSCrypt-proxy could also be configured to do so if you make it your sole DNS server for your network.

To make Unbound DNS work with DNSCrypt-proxy, go to Services > Unbound DNS > General. Since DNSCrypt-proxy supports DNSSEC, you can check “Enable DNSSEC Support” box if you do not already have it enabled for your existing DNS servers. I do not think this option is necessary since Unbound and DNSCrypt-proxy are running on the same machine. The most important thing is that you need to add the following section to your “Custom options” text box:

```
do-not-query-localhost: no
forward-zone:
name: "."
forward-addr: 127.0.0.1@5300
```

Make sure you are using the same port number in the “forward-addr” as you configured for your DNSCrypt-proxy. This will forward all of the DNS lookups from Unbound DNS from all of the devices on your network and send them to the DNSCrypt-proxy. You need to have DNSCrypt-proxy enabled and running before you save your changes because you will now be attempting to use the DNSCrypt-proxy service.

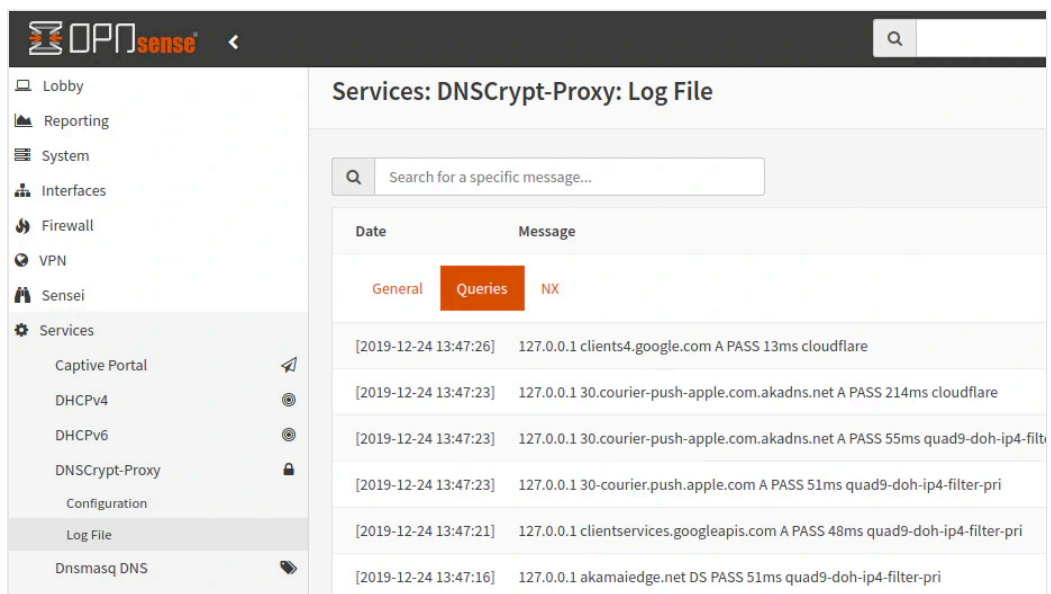


Verify DoH is Configured Properly

Once I was able to get everything up and running, I wanted to verify that it is actually encrypting all of the DNS lookups. There are a number of ways you may try verifying DoH is configured properly. Many of these are described by the [DNSCrypt-proxy wiki](#).

View the DNSCrypt-Proxy Logs

There is a setting called “Fallback Resolver” that allows unencrypted DNS queries, and I was afraid at first that if the DoH configuration was not working that it would fallback to using an unencrypted connection. The help information says it is only used get the IP address of the DNS resolver list. Upon reading more details on the [GitHub page](#), it says the fallback server will not be used for any other queries so one should not have to worry about DNS queries leaking to the unencrypted resolver. Assuming that is accurate, you should be able to know if the DNS queries are encrypted if the results of the DNS queries are showing up in the DNSCrypt-proxy logs by going to Services > DNSCrypt-Proxy > Log File.



Date	Message
[2019-12-24 13:47:26]	127.0.0.1 clients4.google.com A PASS 13ms cloudflare
[2019-12-24 13:47:23]	127.0.0.1 30.courier-push-apple.com.akadns.net A PASS 214ms cloudflare
[2019-12-24 13:47:23]	127.0.0.1 30.courier-push-apple.com.akadns.net A PASS 55ms quad9-doh-ip4-filt
[2019-12-24 13:47:23]	127.0.0.1 30-courier.push.apple.com A PASS 51ms quad9-doh-ip4-filter-pri
[2019-12-24 13:47:21]	127.0.0.1 clientservices.googleapis.com A PASS 48ms quad9-doh-ip4-filter-pri
[2019-12-24 13:47:16]	127.0.0.1 akamaiedge.net DS PASS 51ms quad9-doh-ip4-filter-pri

Use SSH to Resolve a Hostname using the DNSCrypt-Proxy Command

If you are not satisfied with looking at the DNSCrypt-proxy logs, you could take another approach. In the example provided, I used Cloudflare as my DoH service provider. They provide some [documentation](#) on how to verify that DNSCrypt-proxy is running properly. Their example is written generically for those using DNSCrypt-proxy as a client that is installed on any supported platform. The same DNSCrypt-proxy client is used in OPNsense so the same concepts should apply.

If you log into your OPNsense router via SSH, you can run the following command:

```
dnscrypt-proxy -resolve cloudflare-dns.com
```

You should receive an output similar to the following:

```
Resolving [cloudflare-dns.com]

Domain exists:  yes, 3 name servers found
Canonical name: cloudflare-dns.com.
IP addresses:   2400:cb00:2048:1::6810:6f19, 2400:cb00:2048:1::6810:7019,
104.16.111.25, 104.16.112.25
TXT records:   -
Resolver IP:   172.68.140.217
```

The documentation does not state this, but I am assuming that an incorrect configuration would result in resolution of the domain name to fail. I am also assuming that if it resolves correctly, it is correctly using DNS over HTTPS to resolve the domain name and not using an unencrypted DNS service instead. I wish the documentation was more clear so that I can be assured DoH is configured properly.

I have noticed that if I try testing if DoH works with Cloudflare using their [test site](#), it does not seem to properly detect that DoH is configured properly. I am thinking it is due to [how I have my DNS set up](#) in my home network. Since I am using [Pi-hole](#) which forwards to Unbound DNS on my OPNsense router and Unbound forwards to DNSCrypt-proxy which is also on my OPNsense router, I suspect that it may seem to Cloudflare that I am not using DoH because by the time the DNS response reaches my devices, it is no longer encrypted on my internal network. The clients in my network are not getting a direct response back from the Cloudflare DNS servers like it would if the web browser itself is configured to use DoH.

To test this theory, I set up Firefox to use DoH and it passed the tests. I feel fairly certain that the Cloudflare test will not work for my network-wide DoH configuration since DoH is not used in the internal network – only on the edge of the network on my router. For my purposes, I am ok with DNS being unencrypted internally and only encrypting all external DNS requests. It makes it easier to inspect the DNS traffic going through my network, and I can still benefit from DoH since the external DNS lookups are encrypted. Keep in mind that ISPs will still be able to see the IP addresses of where you visit, but it will be harder for them to track exactly where

you are going since multiple websites can be hosted via the same IP address.

Temporarily Turn Off DNSCrypt-Proxy

One quick way to determine if your DNS lookups are being “leaked” out of your network is to turn off DNSCrypt-proxy after you have it configured properly. When you shut it down, all DNS lookups should fail when trying to open a website, for instance. Once you re-enable it, you should be able to resolve domain names again. If the DNS lookups are being leaked out of your network, you would still be able to access websites, etc. on your network. If that is the case, it means your devices are using an alternate DNS server to perform lookups.

In OPNsense, you can stop a service from running by going to the Lobby > Dashboard. You should be able to see that the DNSCrypt-Proxy service is running and be able to click the “Stop” button. You can also click the “Play” button to start the service again.

Packet Capture and Examine Outgoing Packets using Wireshark

One last thing you could try if you are super paranoid and/or more technical is to do a packet capture to ensure no DNS port 53 traffic is leaking out of your network. The only exception of unencrypted DNS traffic may be the fallback DNS server that is specified in the DNSCrypt-proxy settings, but that should only be used to get the IP’s of other DNS resolvers. However, if that is happening, it could indicate that there is some configuration issue since the documentation states that the fallback server is only used when the normal means for obtaining the IP addresses of the DNS resolvers fail. OPNsense provides a mechanism to do a packet capture from the web GUI, but you will need an application like Wireshark to analyze the packets. At the time of this writing, I have not attempted to do so, but I wanted to mention this possibility for those who are technically curious (but if you are that technical, you likely do not need me to mention this possibility).

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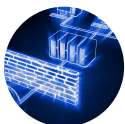
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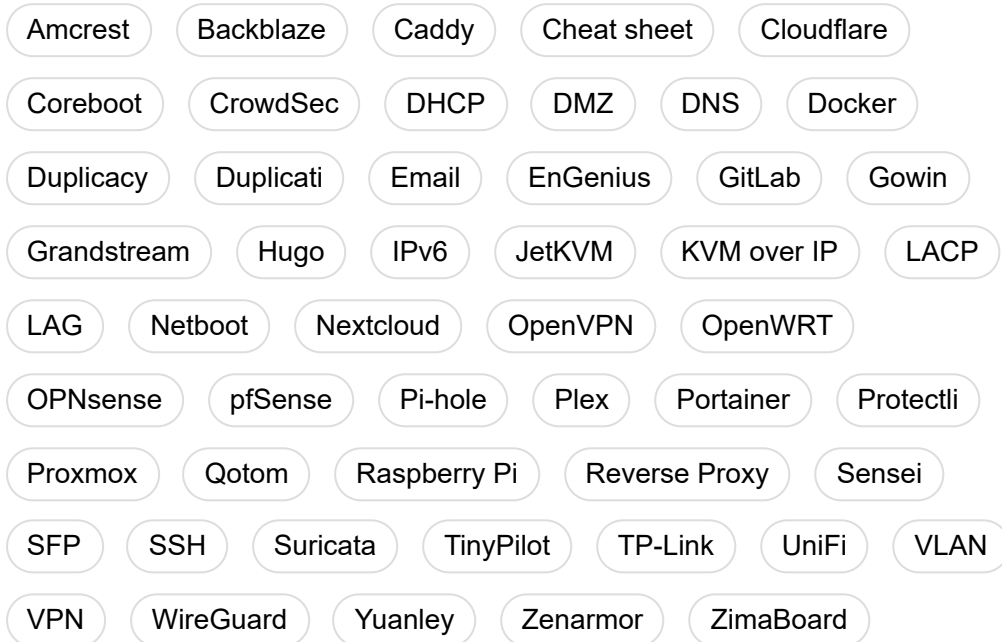
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Misael Rosario

6 years ago

Nice tutorial, I just need to know how to add another DNS to an specific IP address on my network. I used a DNS proxy service to watch a few streaming services overseas. I really like DoH, but with that setup I'm unable to use the DNS server I need on my streaming devices. I believe my only option is to have to VLANs, one with Doh and the other with DNS. Thanks in advanced.

0 0 Reply 



Home Network Guy Mod

6 years ago

➔ Misael Rosario

Sorry for the delay, I was a little busy the last few days. If you create a static DHCP mapping in OPNsense for the specific device, you can specify the two DNS servers to assign to the device. You don't necessarily have to use a separate VLAN just for that one device. You could potentially manually assign the IP address/DNS server on the device itself if you don't want to do it in the router, but I think it's nice to do it in the router since you can leave all your devices on automatic assigned addresses and you have a central location for configuration. It's a pain to have to go to each device and configure them manually if you

have a lot of devices.

0 0 Reply 

S

STX

6 years ago

Fantastic...improving sec on DNS and found this very informative and helpful. One thing, if using internally something like FreeIPA can this create clashes on the opnsense FW? I am thinking about shifting from Unbound DNS to FreeIPA in the near future and am concerned about internal/external look-ups not functioning properly, specially DHCP interactions. Cheers!

0 0 Reply 



Home Network Guy Mod

→ STX

6 years ago

Thanks for the positive feedback! You mainly have to worry about conflicts if you are trying to run more than one DNS service on the same box. If you turn off Unbound DNS and point your network clients to another DNS server (via DHCP), everything should work fine. You could still use DNSCrypt-Proxy with another DNS server if you like or you could just use DNSCrypt-Proxy as your primary DNS server (I haven't tried that because I like to use Unbound DNS to have local hostname resolution on my network).

0 0 Reply 

S

STX → Home Network Guy

6 years ago

FreeIPA would be on separate metal due to critical services. Operated both environments (opnsense & IPA) separately but not in combination yet, a lot of overlapping services. Will most likely dhcp relay with the IPA server and go that route. As IPA is also to be the internal DNS resolving local hosts will have to do some hybrid situation in respect to DNSCrypt-Proxy. We will see how it goes....thanks again!

0 0 Reply 



Home Network Guy Mod

→ STX

6 years ago

Nice. You can turn off any duplicate services so you only have one set of services so you shouldn't have to worry about conflicts. I think the set up is easier if they're all on one box but I



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understand that it is nice to run other services on separate boxes for various reasons. DNS is one thing that is good to have on a separate box if it using your router's DNS server since you want it to be up so your your network doesn't go down — a very critical service.

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clients to go through your local DNS servers but that doesn't work if the clients are tunneling encrypted DNS connections. I will put it in another how-to if you are interested.

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C

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→ Home Network Guy

5 years ago

Thank You for the Link! I created yet the custom-options.conf , pasted the information and restarted, and now Unbound is as well up and running. Regarding DNS (Block) or redirect Rule, as i have now the Config with Unbound and DNSCrypt Proxy like in your Article, what would the Rule look like? Something like this:

<https://forum.opnsense.org/...> Thx

0

0

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Home Network Guy

Mod

→ Criss

5 years ago

Yeah I think that should work for the redirection. I wrote about it also in detail if you are interested:

<https://homenetworkguy.com/...>

0 0 Reply 

C

Criss → Home Network Guy

— 

5 years ago

Thanks, i checked your Article, and i must say again, perfect detailed explained! I just saw that there is a difference, as im doing this Setup for my LAN. The Nat Port Forward Rule Destination should be LAN Address or as in your Article (IOT net) , so in my case LAN net ?(LAN Address was in the opnsense forum Link as Destination specified) Another quick question, I have under the LAN Firewall Rules the dns to local rule on top, and when i enable the allow lan to any rule it works, but it did not work when i disable allow lan to any and enable just allow http / https traffic . If i allow outgoing just http / https (IPv4 TCP) do i need than after the dns to local rule additional the Rule LAN net allow Port 53 (UDP) as well? Thank You!

0 0 Reply 



Home Network Guy Mod

— 

→ Criss

5 years ago

I'm not quite sure if you meant to post this under a different page since you are discussing various firewall rules. If you see any of my examples use a certain interface/network, you may simply change it to the interface/network of your choice. You only need to ensure you are adding the rule to the appropriate network.

If you are wanting to allow only HTTP/HTTPS traffic, at a minimum you would need to allow DNS on port 53 (UDP) if you are using standard unencrypted DNS or other ports if you are using DoT or DoH. The destination

